MILITARY SPECIFICATION

VACUUM PUMPS, POWER DRIVEN, CENTRIFUGAL

PUMP PRIMING SERVICE, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers the design and construction of vacuum pumps for the purpose of priming centrifugal pumps by servicing and associated vapors from pump casings and suction piping.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearing, Ball, Annular, (General Purpose).
GGG-P-781 - Pullers, Bushing, Bearing, Gear, Wheel Hub and Cylinder Sleeve, Installing and Removing.

MILITARY

JAN-P-735 - Primer, Paint, Zinc-Chromate Alkyd Type.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-C-2174 - Controllers, Motor Starters, and Master Switches - Direct-Current, Naval Shipboard Use.
MIL-C-2212 - Controllers, Motor Starters, and Master Switches - Alternating-Current Naval Shipboard Use.
MIL-I-15024 - Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-T-15071 - Technical Manuals for Mechanical and Electrical Equipment.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15328 - Coating, Pretreatment (Formula No. 117 for Metals).
MIL-M-16576 - Metal, Gun: Castings.
MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts.
MIL-M-17059 - Motors, Alternating Current, Fractional HP (Shipbpard Use).
MIL-M-17060 - Motors, Alternating Current, Integral HP (Shipboard Use).
MIL-M-17413 - Motor, Direct-Current, Integral HP (Shipboard Use).
MIL-M-17506 - Nickel-Copper-Aluminum Alloy, Wrought.
MIL-P-17545 - Primer, Paint, Shipboard, Alkyd-Red Lead Type No. 116.
MIL-M-17556 - Motor, Direct-Current, Fractional HP (Shipboard Use).

FED. SUP. CLASS
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NAVY DEPARTMENT
General Specifications for Inspection of Material
General Specifications for Machinery:
Section S1-3 - Bolt, Bolt-Studs, Stud-Bolts, Studs, Screws, and Nuts.
Section S1-4 - Welding and Allied Processes.

DRAWINGS

BUREAU OF SHIPS
B-153 - Packing and Gaskets, Standard Application of.
B-176 - Flanges, Composition, Silver Brazing, for Copper and Copper-Nickel-Alloy Tubing and Brazed Pipes, Pressures 100 P.S.I. and Below.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Qualification.- Power driven vacuum pumps furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 Materials.-

3.2.1 All material used in the construction of the pumps and associated equipment shall be of the quality best suited for the purpose intended, and shall be strictly in accordance with the material specifications.

3.2.2 The use of materials differing in properties from those specified for a given application will be considered when the contractor shows the necessity for such substitution, the material proposed is readily obtainable, and provided that the material is satisfactory to the bureau or agency concerned.

3.2.3 Where no specifications are specified herein for the material for certain parts, the material used shall be of a type and quality consistent with that specified for associated parts in the pump. The use or substitution of manufacturer's material shall be subject to approval by the bureau or agency concerned and shall be clearly indicated in a list submitted with the master drawings.

3.3 General design.-

3.3.1 The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. It is the intention of this specification to obtain pumps of such design that they will operate over a long period of years with a minimum of servicing. Where wear or erosion is unavoidable the parts subjected to such wear or erosion shall be of the best materials available for the purpose in order to reduce these detrimental effects to a minimum. The design and construction of all pumps shall be the most compact consistent with the following requirements:

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion.
(d) Economy.

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(e) Satisfactory operation when inclined as follows:
(1) For surface vessels:
a. Up to 15 degrees to either side (permanently inclined).
b. With the ship rolling up to 45 degrees from the vertical to either side.
c. With the ship pitching 10 degrees up and down from the normal horizontal plane.
(2) For submarines:

	Submerged	Snorkel	Surface
Trim (permanently inclined)	30 degrees	7 degrees	7 degrees
Pitch	10 degrees	10 degrees	10 degress
Time for complete pitch cycle	4 seconds	4 seconds	4 seconds
List (permanent)	15 degrees	15 degrees	15 degrees
Rolling	30 degrees	30 degrees	60 degrees
Time of complete roll cycle	12 seconds	12 seconds	8 seconds

Note. - Degrees measured from the vertical to either side for list and roll, and from the normal horizontal plane up or down for trim and pitch.

3.3.2 The priming pump shall be of the centrifugal displacement, wet vacuum type and shall be provided with all necessary fittings and accessories including vacuum relief valve, check valve and similar items required for the satisfactory priming of one or more centrifugal pumps as required by the contract or order.

3.3.2.1 Each priming pump shall be furnished with an automatic valve to prevent the flow of liquid from the centrifugal pump or piping system to the primer.

3.3.2.2 The design of the primer shall be such as to keep metal to metal contact to an absolute minimum in order to reduce wear.

3.3.2.3 Consideration will be given to priming pumps of designs other than the centrifugal displacement unit vacuum type, providing: all other features required herein are retained; the performance is equal to or better than that of the centrifugal displacement wet vacuum type; and the weight, space, and power requirements do not exceed those of the centrifugal displacement wet vacuum type.

3.3.3 Mounting. -

3.3.3.1 Pump units shall be horizontal mounted.

3.3.3.2 When specified (see 6.1), the pump unit shall be mounted on a bedplate which also serves as reservoir for the seal wafer.

3.3.3.3 Special attention shall be given to the design of close-coupled pump to insure that leakage from the pump stuffing box glands cannot enter motor frames or bearing brackets.

3.3.4 Bearings. -

3.3.4.1 Radial and thrust bearings shall be of the ball type.

3.3.4.2 The use of ball bearings shall be subject to the speed limitations of Specification FF-B-171 except where prior approval of the bureau or agency concerned has been granted.

3.3.4.3 Installation. -

3.3.4.3.1 Bearings shall be installed in housings separate from and independent of the stuffing boxes and with adequate protection from gland leakage. Suitable means shall be provided to prevent escape of lubricant around the shaft. All baffles, wipers, and related parts shall be readily renewable.

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3.3.4.4 Lubrication of the ball bearings shall be in accordance with Specification FF-B-171.

3.3.5 Motors. -

3.3.5.1 All motors for driving pumps shall be of the types and characteristics specified in the contract or order, and shall conform to Specification MIL-M-17413 or MIL-M-17556 for direct current (d. c.) motors, or Specification MIL-M-17059 or MIL-M-17060 for alternating current (a. c.) motors.

3.3.5.2 The horsepower rating of each motor shall be not less than the maximum brake horsepower of the driven pump under any condition from shut-off to free delivery. The actual motor rating shall be in accordance with Navy standard motor ratings. If the maximum brake horsepower of the pump is less than 2, and if there is a possibility of an unpredictably high frictional load due to improper adjustment of the gland or to some other cause, then the maximum brake horsepower shall be multiplied by a safety factor of 1.5 - $\frac{\text{(max. BHP)}}{4}$ before selecting the next larger Navy standard rating.

3.3.6 Controllers. - All motor controllers shall be of the characteristics specified (see 6.1) and shall conform to Specification MIL-C-2174 for d. c. controllers or Specification MIL-C-2212 for a. c. controllers.

3.3.7 Shockproofness. -

3.3.7.1 The design of all complete pumping units shall be such that they are capable of passing the class HI (high-impact) shock test specified in Specification MIL-S-901.

3.3.7.2 Shockmounts are not to be used unless bureau or agency approval of the mount and its proposed application is given.

3.3.7.3 The determination of stresses resulting from the shock tests is accomplished by the application of the usual methods of analytical mechanics. The allowable stress in parts due to shock shall be the yield strength (0.2 percent offset) of the material. Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses. It is of paramount importance that the maximum accelerations possible under the shock tests be utilized in the design of holding down bolts, supporting feet or lugs, and main structural members of the equipment, and that no failure or permanent deformation occur as a result of shock. Parts of an assembly mounted on a common rigid base need not be designed for higher accelerations than the values for which the whole assembly is designed unless the designer has reason to believe that higher accelerations will be experienced by the parts.

3.3.7.4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance, as experience indicates that large clearances allow impacting and subsequent failure.

3.3.8 Painting. -

3.3.8.1 For pumps operating at a temperature of 300° Fahrenheit (F.) or less, all external unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment, in accordance with Specification MIL-C-15328, and one coat of zinc chromate primer in accordance with Specification JAN-P-735, or one coat of red lead primer in accordance with Specification MIL-P-17545, followed by a finish coat of light gray equipment enamel, in accordance with Specification MIL-E-15090.

3.3.8.2 Painting of external surfaces of nonferrous parts of pumps will not be required.

3.3.9 Threaded parts such as bolts, studs and nuts shall conform to Section S1-3 of the General Specifications for Machinery and Handbook H28.

3.3.10 Welding shall be in accordance with Section S1-4 of General Specifications for Machinery.

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3.4 Detail requirements. -

3.4.1 Casings. -

3.4.1.1 The casings of all pumps shall be designed so as to permit ready removal of the rotors and replacement of bearings and glands without dismounting the unit.

3.4.1.2 Sufficient clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools.

3.4.1.3 Fitted bolts or dowel pins shall be provided to insure maintenance of alignment of casing parts in reassembly.

3.4.1.4 Casings shall be provided with drain connections.

3.4.2 Suction and discharge connections. -

3.4.2.1 Suction and discharge connections shall be flanged in accordance with Drawing B-176.

3.4.3 Rotors and shafts. -

3.4.3.1 Rotor surfaces shall be smooth finished.

3.4.3.2 Rotors shall be keyed on the shaft and securely held against lateral movement by locked nuts or other means approved by the bureau or agency concerned.

3.4.3.3 Shaft threading shall be counter to the direction of rotation except where both right hand and left hand units of the same design are involved, in which case, in order to reduce the number of shafts required for repair parts, the threading shall be right hand for all units and the threaded parts shall be locked in place.

3.4.3.4 Shafts shall be fitted with shaft sleeves in way of stuffing boxes.

3.4.3.5 The shafts of all pumps, except those driven by a. c. motors, shall be provided with means to permit the ready and direct arc of portable tachometers.

3.4.4 Stuffing boxes and packing. -

3.4.4.1 Stuffing boxes shall be of adequate depth and design to reduce leakage to a minimum under all operating conditions.

3.4.4.2 Packing shall be in accordance with Drawing B-153. All packing shall be of a brand approved by the bureau or agency concerned.

3.4.4.3 The design shall be such as to insure that leakage from the glands cannot reach the bearings or be thrown over bearing housings or driving units. This shall preferably be accomplished by using glands cored out internally and the shafts provided with flingers. Ample drip pockets, spray shields and drains shall be provided.

3.4.4.4 Glands shall be split in halves with the halves secured together in a manner to form solid rings for setting up.

3.4.4.5 All stuffing box glands shall be set up by units threaded on gland swing bolts, or on studs secured in the casings.

3.4.4.6 Sufficient space shall be provided between bearings and stuffing boxes to permit easy removal of packing and inspection and overhaul of bearings.

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3.4.5 Couplings. -

3.4.5.1 Flexible couplings, when required, shall be of all metal construction.

3.4.5.2 All flexible coupling hubs shall be keyed to the shafts and secured by locknuts. For shafts 1 inch in diameter and larger the hubs shall be fitted on a taper with keys parallel to the tapers.

3.4.6 Priming valve. -

3.4.6.1 Priming valves shall be of the float type and shall permit the flow of air while the pump is being primed, shall close when the water level reaches the valve to prevent water from entering the priming pump, and shall open whenever the centrifugal pump loses its prime.

3.4.6.2 Inlet and outlet connections shall be flanged in accordance with Drawing B-176.

3.4.7 Seal water tanks. -

3.4.7.1 The seal water tank shall be of welded construction with suitable reinforcements to accommodate the weight of the priming pump unit.

3.4.7.2 The tank shall be fitted with a water gauge glass complete with shut-off valves, filling connection, drain connection, vent and overflow connection, and hand hole.

3.4.8 Materials. - Materials for the pump shall be in accordance with table I:

Table I - Material.

Name of part	Material	Specification
Casing	Gun metal	MIL-M-16576
Rotor	Gun metal	MIL-M-16576
Gland	Gun metal	MIL-M-16576
Shaft	Ni-Cu-Aℓ alloy	MIL-N-17506
Shaft sleeve	NI-Cu-Aℓ alloy	MIL-N-17506
Mounting brackets	Gun metal	MIL-M-16576
Seal water tank	Galvanized steel	------
Priming valve casing	Gun metal	MIL-M-16576
Priming valve float	Copper, rubber	------

3.4.9 Identification plates. -

3.4.9.1 Identification plates shall be furnished for each pump and shall be type A or B in accordance with Specification MIL-I-15024, except that identification plates of plastic or aluminum are not acceptable.

3.4.9.1.1 A separate identification plate shall be furnished for the priming valve.

3.4.9.2 Identification plates shall be secured to equipment with corrosion resistant metallic screws.

3.4.9.3 Pump identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Service application (that is, vacuum priming pump).
(d) Manufacturer's serial number.

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(e) Salient design characteristics.
  (1) Capacity in cubic feet per minute.
  (2) Vacuum in inches of mercury.
  (3) Speed in revolutions per minute.
  (4) Brake horsepower.
(f) Contract number.
(g) Bureau of Ships stock number.
(h) Section for Government inspector's stamp.

3.4.9.4 Priming valve identification plate shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or size.
(c) Manufacturer's serial number.
(d) Service application (that is, vacuum priming valve).
(e) Contract number.
(f) Bureau of Ships stock number.
(g) Section for Government inspector's stamp.

3.4.9.5 Each driving unit and accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.5 Drawings.-

3.5.1 Drawings shall be in accordance with type I of Specification MIL-D-963.

3.5.2 Outline drawings shall be furnished covering complete pump units including pump, driver, bedplate and attached accessories. Separate outline drawings shall be furnished for each contract or order and for each different pump unit on a contract or order.

3.5.2.1 Performance curves on preliminary outline drawings as orginally submitted for approval to the bureau or agency concerned may be calculated curves and shall be so indicated. The complete performance curves on the validated outline drawings shall be based on the performance acceptance tests and shall be so indicated.

3.5.3 Pump master drawings shall include sectional assembly, complete list of material and details of all parts. The section assembly and list of material with references to detail drawings should be on a single sheet; and details shall be on one or more additional sheets. Several details should be shown on each pump master drawing. Details of single pump parts on size A sheet, resulting in a multiplicity of drawing sheets, should be avoided.

3.5.4 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "List of Material" of the sectional assembly drawing.

3.5.5 Drawings for drivers and associated equipment shall be in accordance with the specifications for those equipments.

3.5.6 The sectional assembly drawing shall show all assembly clearances.

3.5.7 Drawings submitted with bids are not required to be in master drawing format.

3.6 Technical manuals.- Technical manuals shall be in accordance with type C of Specification MIL-T-15071, unless otherwise specified in the contract or order. A separate technical manual shall be furnished for each different pump unit except in special cases when manuals covering more than one pump may be approved. Preliminary technical manuals submitted for approval shall include all proposed sections complete.

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3.7 Repair parts and tools. -

3.7.1 Onboard repair parts of the types and quantities shown in table II, plus any additional parts considered necessary to properly maintain the pump shall be furnished as required in the contract or order.

Table II - Onboard repair parts.

Name of part	Quantity per set
Rotor	1
Rotor key	1
Shaft sleeve	1
Packing	1 set
Water flingers and set screws	1
Casing (excluding mounting bracket	1
Gaskets	1 set
Locknuts and washer for rotors	1
Coupling wearing parts	1 set
Priming valve float	1
Priming valve bushing	1
Priming valve seat	1
Priming valve gaskets	1 set
Relief valve	1

(When flexible coupled or belt driven units are furnished, one complete rotating assembly should be furnished in lieu of the duplicate parts listed herein. The coupling or sheave should be included.)

3.7.1.1 Repair parts for the drivers and associated equipments shall be furnished in accordance with the applicable equipment specification. Motor rotors or armatures shall be furnished for close-coupled units.

3.7.1.2 A complete set of special tools required for the maintenance and repair of the pump units shall be included in each set of onboard repair parts.

3.7.1.2.1 Tools which are available in the Catalog of Naval Material, General Stores Section, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with Specification GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

3.7.1.2.2 Each box containing onboard repair parts shall contain a list entitled "List of onboard repair parts and tools". The list shall be in a format suitable for use on outline drawings and in the technical manuals. It shall not be less than nominal 8-1/2 by 11 inches in size.

3.7.1.2.2.1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entrees as follows:

Reproduced from drawing number _____
Number of ships _____
Application _____
Contract number _____
Manufacturer _____
Quantities are for _____ units per ship

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The columns shall include:

Piece number _____
Name of part _____
Quantity _____
Drawing number _____
Standard Navy stock number _____

Additional columns may be used as applicable.

3.7.1.2.2.2 The list shall be so treated as to be resistant to oil, water and fading.

3.7.2 Stock.- The selection, stock numbering and quantities of repair parts for stock shall be determined and processed in accordance with Specification MIL-R-15137.

3.7.2.1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania.

3.8 Workmanship.- The equipment, including all parts and accessories, shall be manufactured and finished with first class workmanship in all respects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests at a Government laboratory.- Qualification tests shall be conducted at the Engineering Experiment Station, Annapolis, Maryland. The characteristics of the units for qualification tests shall be as specified by the Bureau of Ships.

4.1.1 The qualification tests shall adequately demostrate the ability of the pump to handle the specified capacity at the specified vacuum. Sufficient data shall be taken during the tests to prepare pump characteristic curves as required herein. Test data shall be complete over the entire range of capacities from shut-off to as near free delivery as possible. The following test curves shall be prepared.

(a) Vacuum versus air capacity.
(b) Vacuum versus brake horsepower.

4.1.2 Endurance test.- The pump shall be operated for 24 hours continuously at a capacity as a near free delivery as possible and for 12 hours at a capacity as near shut-off as possible. Failure of any part of the pump during the endurance test shall be considered as failure of the pump to qualify. The pump shall complete the endurance test without excessive noise, vibration or wear of parts.

4.1.3 Consideration will be given to granting qualification as a result of service tests. To be granted qualification as the result of service tests, the following requirements shall be met:

(a) Two or more pumps of the general design and general operating characteristics shall have given satisfactory service onboard a Naval vessel for a period of at least two years and shall still be in service on such vessel at the time qualification is requested.
(b) When the design of a pump in service for which a manufacturer requests qualification does not meet current Naval requirements as regards vacuum and capacity characteristics, the manufacturer shall be required to furnish satisfactory evidence that he has produced pumps of a design, and characteristics equal to those currently required for Naval service. The Bureau of Ships reserves the right to require additional qualification tests of any design of pump, which, in its opinion, has been changed sufficiently to require further demonstration of its suitability.
(c) Any manufacturer whose product has been eligible to be placed on the qualified products list as a result of service test shall be required on his first contract or order after being granted qualification to submit a complete unit to the Engineering Experiment Station, Annapolis, Maryland, for a check test which will consist of the qualification tests. The awarding of qualification shall be contingent upon the unit passing the check test.

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4.2 Shop tests. -

4.2.1 Each part shall be tested hydrostatically to a pressure of at least 150 percent of the maximum pressure to which it can be subjected in operation, but not less than the pressures listed as follows:

Pump casings	50 p.s.i.g.
Priming valve body	30 p.s.i.g.
Priming valve float	25 p.s.i.g.
Seal tank	50 p.s.i.g.

4.2.2 Each pump, except those driven by a.c. motors, shall be tested in the shop of the manufacturer or contractor at 25 percent above designed maximum operating speed by a continuous nonstop run of at least 30 minutes. This test shall be conducted in the presence of the Government inspector, who shall check operation and smoothness of running. In the case of pump driven by a.c. motors each unit shall be tested for smoothness of operation at full speed.

4.3 Performance acceptance tests. -

4.3.1 It shall be the responsibility of the prime contractor to insure that tests required on assembled units are made.

4.3.2 Driving units shall be tested as required by the applicable specification.

4.3.3 Performance acceptance tests shall adequately demonstrate the ability of the pump to handle its rated capacity of air at the specified vacuum.

4.3.4 Sufficient data shall be taken during the tests to prepare pump charateristic curves as specified in 4.3.5. Test data and curves shall be complete over the entire range of capacities from shutoff (highest vacuum obtainable) to as near free delivery (atmospheric pressure) as possible.

4.3.5 The following test curves shall be supplied for constant speeds of approximately 90, 100 and 110 percent of designed speed:

(a) Vacuum versus air capacity.
(b) Vacuum versus capacity in free air.
(c) Vacuum versus brake horsepower.

Where pumps are driven by a.c. motors, the cuvres shall be obtained for the rated speed conditions of the motor in lieu of those noted above.

4.3.6 Complete test reports, including test data on both pumps and driving units, shall be assembled and distributed as follows:

(a) Three to the Bureau of Ships (to be forwarded via the cognizant Government inspector).
(b) One for primary district inspection office.
(c) One for branch inspection office (if any).
(d) Two to each supervisor of shipbuilding (one for supervisor's files, the other for the shipbuilder).
(e) One to the Commander of each ship building yard concerned, for Navy yard built ships, when specified in the contract or order.

4.3.7 Shock tests. -

4.3.7.1 One complete pumping unit of each type, design and size shall be subject to the high impact shock test specified in Specification MIL-S-901 as required by 3.3.7.

4.3.7.2 Equipment previously accepted will not be resubmitted for shock test except when evidence of low shock resistance develops in the units installed. The bureau or agency concerned will initiate action with the manufacturer for correction of deficiencies, and reserves the right to require shock tests at any time.

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4.3.7.3 All pump units shall be shock tested with drivers unless otherwise approved by the bureau or agency concerned. Flexible coupled pumps shock tested with one driver will not be required to be shock tested again when supplied with a different driver. Prime movers are subject to shock tests in accordance with the applicable equipment specification.

4.3.7.4 The pump unit shall be considered to have failed to pass the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part, such as shaft, mounting feet, bearings.
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to tests at rated speed, or more than that permitted by the specifications for the driver.

4.3.7.5 Shock tests shall be conducted at the manufacturer's plant. Where a manufacturer is unable to conduct these tests at his own plant, he may arrange to have them conducted at a commercial laboratory, or Government laboratory suitably equipped to conduct tests. Where shock tests are conducted at a Government laboratory, copies of the applicable master drawings shall accompany the units.

4.3.7.6 Pump units which have been subjected to the high-impact shock test and have failed to conform to the requirements will not be acceptable.

4.3.7.7 Pump units which have been subjected to the high-impact shock test and have successfully passed this test shall be considered acceptable for such service as the bureau or agency concerned may authorize. Such equipment will be returned by the Government to the contractor for reconditioning as follows:

(a) Minor deformations of mounting flanges shall be corrected.
(b) Minor deformations affecting alignment shall be corrected.
(c) All bearings shall be replaced.
(d) Each part shall be carefully examined by the contractor and any part which he considers substandard shall be replaced.

Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high-impact shock test but which are in excess of the design dimensional tolerances specified on the applicable pump drawings.

4.4 Inspection of pump repair parts. - The pump repair parts required by 3.7.1 shall be inspected to assure interchangeability with the similar parts in the assembled pumps. Ordinarily material analyses and physical test specimens will not be required but the Government inspector shall employ such procedures as will assure the delivery of parts made of the same materials and by the same processes as for the assembled pumps.

4.5 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. - Pumps shall be prepared for delivery for domestic shipment - immediate use; domestic shipment and storage; or for overseas shipment in accordance with Specification MIL-P-16789 (see 6.1).

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Rated capacity (c.f.m.).
(c) Vacuum (in. Hg.).
(d) Mounting required (see 3.3.3.2).
(e) Whether seal tank or valve is required.
(f) Liquid handled by centrifugal pump if other than water.
(g) Whether close coupled or flexible coupled to driver.
(h) Maximum pressure to which priming valve will be subjected.
(i) Electrical characteristics (for motor drive) as required by applicable motor and controller specifications.
(j) Whether shipment is for domestic shipment - immediate use; domestic shipment and storage; or for overseas shipment (see 5.1).
(k) If repair parts boxes are required, specify type (metal or wood) (see 5.1).

6.1.1 Bid data. - Bid data shall include the following data:

(a) Sectional assembly drawing of pump.
(b) A drawing of the unit showing overall dimensions.
(c) Pump characteristic curves.
(d) Pump efficiency at rated capacity.
(e) Horsepower of driver.
(f) R.p.m. at rated capacity.
(g) Weight of complete unit.
(h) The manufacturer and descriptive data of the driven and associated accessories, such as controller.

6.2 Qualification. - In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C.

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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